WORKED SOLUTION

STEP II 2016, Question 9

Mechanics • Momentum and Energy Loss

STEP II 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

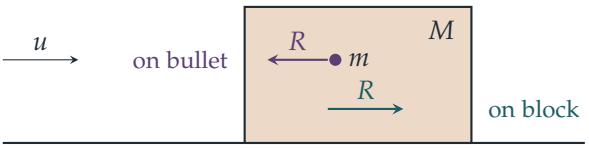
step.maths.org/questions/16-s2-q9

The key idea

The resistance R is an internal force between bullet and block: it decelerates the bullet and, by Newton's third law, pushes the block forward with the same magnitude. So momentum of the pair is conserved, while kinetic energy is not. The energy lost is exactly the work done by R through the *relative* distance the bullet moves inside the block. Applying the work–energy principle to the block alone gives how far the block moves.

Solution

We model the bullet and block as particles moving along one horizontal line, with the resistance R constant while the bullet moves relative to the block. Take the bullet's initial direction as positive.



The vertical forces (weights, normal reaction from the table) balance and play no part. Horizontally, the bullet feels R backwards and the block feels R forwards.

Part (i)

With the block fixed, the bullet has constant deceleration R/m. Using $v^2 = u^2 + 2\alpha s$ with final speed 0 and $\alpha = -R/m$:

$$0 = u^2 - \frac{2R}{m}a \quad \Longrightarrow \quad \boxed{a = \frac{mu^2}{2R}.}$$

(Equivalently, the work done against the resistance, Ra, equals the bullet's initial kinetic energy $\frac{1}{2}mu^2$.)

Part (ii)

Common final velocity. No external horizontal force acts on the system of bullet and block, because the table is smooth. When the bullet is at rest relative to the block, both move with the same velocity V, and conservation of momentum gives

$$mu = (M+m)V \quad \Longrightarrow \quad V = \frac{mu}{M+m}.$$

Ask yourself The resistance acts through a distance b on the bullet's path inside the block, but the bullet itself travels $b+c$ over the table. Which distance goes into the energy equation?

Finding b. Relative to the table, the force R on the bullet does work $-R(b+c)$, since the bullet moves $b+c$; the force R on the block does work $+Rc$, since the block moves c. The net work done on the system is therefore $-Rb$: only the relative displacement counts. By the work–energy principle,

$$\begin{aligned} Rb &= \tfrac{1}{2}mu^2 - \tfrac{1}{2}(M+m)V^2 \\ &= \tfrac{1}{2}mu^2 - \frac{m^2u^2}{2(M+m)} \\ &= \tfrac{1}{2}mu^2\left(1 - \frac{m}{M+m}\right) = \tfrac{1}{2}mu^2 \cdot \frac{M}{M+m}. \end{aligned}$$

Since $\frac{1}{2}mu^2 = Ra$ from part (i), dividing by R gives

$$\boxed{b = \frac{M}{M+m}\,a.}$$

Finding c. Now apply the work–energy principle to the block alone. It starts at rest, ends with speed V, and the only horizontal force on it is R forwards, acting through distance c:

$$Rc = \tfrac{1}{2}MV^2 = \frac{Mm^2u^2}{2(M+m)^2} = \tfrac{1}{2}mu^2 \cdot \frac{mM}{(M+m)^2}.$$

Using $\frac{1}{2}mu^2 = Ra$ again,

$$\boxed{c = \frac{mM}{(M+m)^2}\,a.}$$

An alternative via kinematics. The bullet decelerates at R/m and the block accelerates at R/M. Their velocities agree when $u - Rt/m = Rt/M$, so

$$t = \frac{mMu}{R(M+m)}, \qquad c = \frac{1}{2}\cdot\frac{R}{M}\,t^2 = \frac{m^2Mu^2}{2R(M+m)^2},$$

which agrees with the above, and b follows from the bullet's displacement minus c.

Checks

- As $M \to \infty$ the block behaves as if fixed: $b \to a$ and $c \to 0$, as they should.
- Since $\frac{M}{M+m} < 1$, we have $b < a$: a movable block absorbs less of the bullet's energy as heat, so the bullet does not penetrate as far.
- With $M = 4m$ we get $b = \frac{4}{5}a$ and $c = \frac{4}{25}a$. Stepping the motion forward numerically in small time steps, for several random choices of m, M, u and R, reproduces a, b and c to four significant figures.

Common slips

- Using the bullet's distance over the table, $b + c$, in the energy-loss equation instead of the relative distance b.
- Assuming kinetic energy is conserved in the collision; it is not, and the loss Rb is the heat generated.
- Forgetting that the block feels the force R too (Newton's third law), and so leaving the block's motion unexplained.

Beyond STEP

The fact that internal forces do zero net work only when there is no relative motion is the start of a careful treatment of energy in systems of particles. Working in the centre-of-mass frame makes it clean: the kinetic energy available for dissipation is $\frac{1}{2}\mu u^2$, with reduced mass $\mu = mM/(M+m)$, which is exactly Rb here. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas thoroughly.

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